

Supercritical Carbon Dioxide cycle with precompression and recuperation

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Abstract

This document provides a detailed overview of the Supercritical CO₂ Power Cycle, explaining its operation and the selection of key thermo-fluid devices like turbines and heat exchangers. The report presents a study on a specific configuration of this cycle (Supercritical CO₂ Power Cycle with Precompression and Recuperation), considering parameters and utilizing a computer code for analysis. The model achieves a thermal efficiency of approximately 39.16% at a turbine inlet pressure of 36.5 MPa, aiming for a power capacity of around 150 MW through multiple iterations. Additionally, a thermo-economic analysis was conducted for the Power Cycle. This involved modelling the Low Temperature Recuperator (LTR) and performing economic assessments, including component-wise costs and overall capital costs for the power plant. The results were obtained through plant-level simulation using Python and CoolProp, providing valuable insights into the performance and cost aspects of the Supercritical CO₂ Power Cycle with Recuperation.

Keywords: Supercritical carbon dioxide, LTR, power cycle, thermal efficiency, Precompression, thermo-economic

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Introduction

S-CO₂ cycle with pre-compression and recuperation uses sCO₂ as the working fluid (shown in figure 1.). It is a simple Brayton cycle which operates at high pressures hence resulting in higher efficiencies. Because of higher efficiencies, various models of this cycle are in development. CO₂ being a fuel which is eco-friendly is another reason why this cycle is in development.

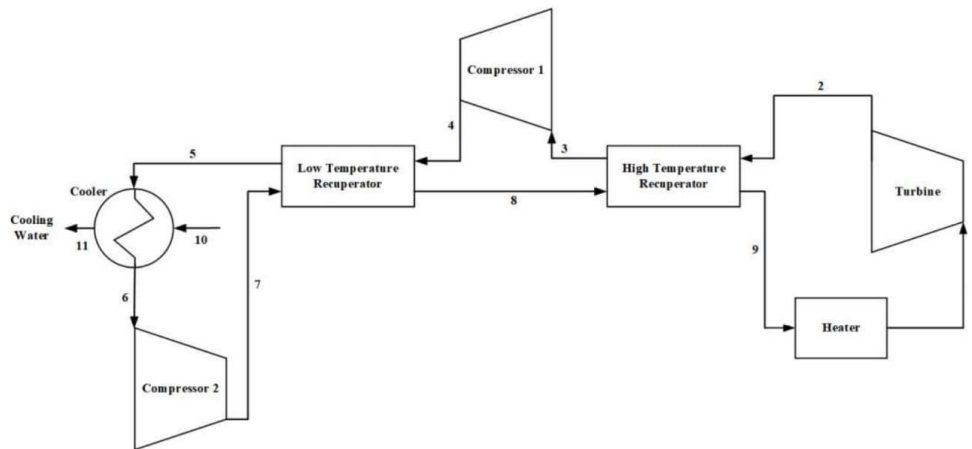


Figure 1. sCO₂ Power Cycle with Precompression

Each process in the cycle is explained as follows:

1 to 2 - Isentropic(ideally) expansion of sCO₂ in a turbine

2 to 3 - The fluid then passes through High Temperature Recuperator (HTR) where the heat is lost from the fluid coming out of the turbine and that heat is used to heat up the working fluid entering the heater

3 to 4 - Isentropic(ideally) compression of sCO₂ in the pre-compressor

4 to 5 – Working fluid goes through LTR where the heat is again lost from the fluid coming out of the pre-compressor and that heat is used to heat up the working fluid entering the HTR

5 to 6 - The fluid is then cooled in the cooler. The working fluid of this cooler is water.

6 to 7 - Isentropic(ideally) compression of sCO₂ in the main compressor

7 to 8 - Fluid enters LTR where it gains some heat

8 to 9 - Fluid then enters HTR where its temperature is further increased

9 to 1 – The fluid is heated in a conventional coal-fired heater to raise its temperature

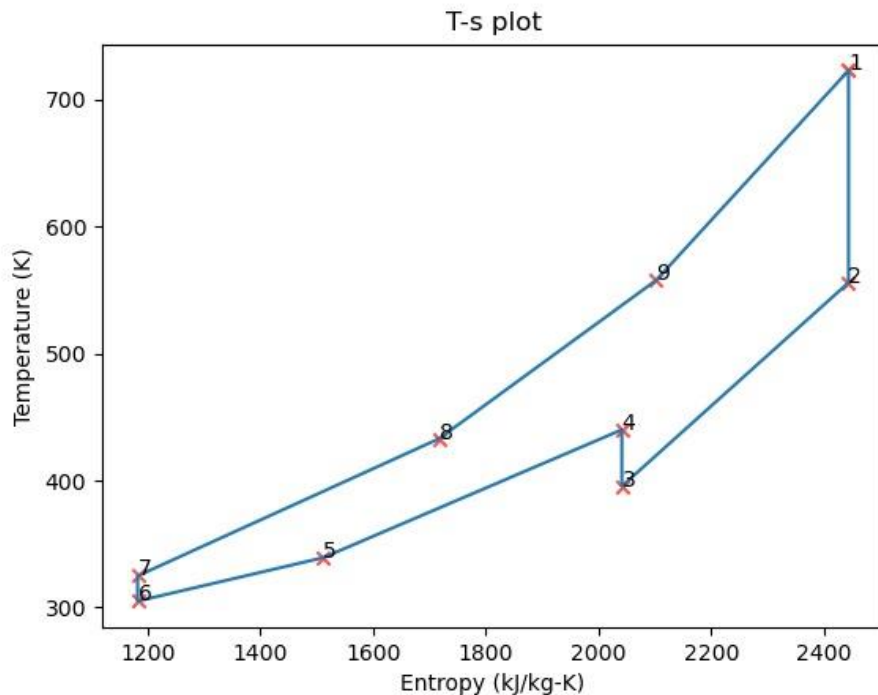


Figure 2. sCO₂ Power Cycle T-s Diagram

In our plant, the s-CO₂ cycle consists of a compressor, heater, turbine, pre-compressor, main compressor, HTR, LTR, and a cooler. We use HTR and LTR to decrease the amount of power required for the heater and cooler to heat and cool the fluid respectively. We are using coal-fired heater. The power plant is designed to provide 150 MW.

We now proceed to choose various operational parameter, perform a simulation of the plant and analyse the efficiency and costs associated with it.

Methodology

Choice of components

The power outlet rating of the plant is 150 MW, accordingly all the components were chosen.

Main compressor and Pre-Compressor: The compressor we choose for our system is an axial flow compressor. Axial flow compressors are particularly well-suited for s-CO₂ power cycles because they can operate near the critical point of CO₂, where the fluid properties are unique and challenging for other types of compressors. Near the critical point, the viscosity of CO₂ is very low, which makes it easier to compress and reduces the amount of work required. The turbine and all compressors are running on the same shaft.

Turbine: The turbine we choose is also multistage axial flow turbine, multi-stage axial flow turbines offer a compelling combination of efficiency, compactness, adaptability, and reliability, making them the preferred choice for SC-CO₂ power cycles. Their ability to handle the unique properties of SC-CO₂ and deliver high power output makes them a valuable technology for next-generation power generation systems.

Cooler: The cooler has to cool the fluid to a very less extent due to the presence of LTR and HTR. Shell and tube type heat exchanger works well in our case having water in shell side and s-CO₂ in tube side.

Heater: We employed combustion type aka Direct-fired heaters (DFHs). In DFHs, the combustion of fuel takes place directly within the heat exchanger, allowing for direct heat transfer to the CO₂. DFHs offer high heating rates and can achieve high temperatures, making them suitable for applications where rapid heating is required.

LTR and HTR: The LTRs and HTRs we used are micro channel heat exchangers (MCHE) type. MCHEs offer several significant advantages for use as recuperators in s-CO₂ power cycles. Their high heat transfer coefficient, compact size, high efficiency, improved thermal-hydraulic performance.

Calculation of Thermodynamical parameters in the plant

Assumptions

Some basic assumption which we took under consideration are: -

- Pressure change occurs only in turbine, pre-compressor and compressor, and there is no pressure drop in any of the heat exchangers.
- Turbine, compressor and pre-compressor are isentropic in nature.
- We fixed turbine compressor ratio and main compressor ratio using literature [1], [4] and found out pre-compressor ratio using equation (0).

Assumed values and design specifications are given in the table below: -

Table 1: Standard design conditions of the proposed system / Assumed Parameters

S. No.	Design Parameter	Value
1	Turbine inlet temperature	450°C
2	Turbine inlet pressure	36.5MPa
3	Turbine efficiency	0.9
4	Turbine pressure ratio	4.0
5	Pre-compressor outlet temperature	440°C
6	Pre-compressor efficiency	0.86
7	Compressor efficiency	0.86
8	Compressor pressure ratio	2.5
9	Effectiveness of HTR	0.9
10	Effectiveness of LTR	0.9

Process to calculate plant parameters

Notation:

P_i : Pressure at point i

T_i : Temperature at point i

s_i : Specific entropy at point i

h_i : Specific enthalpy at point i

Note: The values if i can be referred from figure 1.

1. We started our study of the cycle by assuming the values of Turbine inlet temperature (T_1) and pressure (P_1) of $s\text{CO}_2$ and determining the value of enthalpy (h_1) of $s\text{CO}_2$ at that point using CoolProp[4]
2. Then we found pressure (P_2) at outlet of the turbine using equation (1)
3. As we have considered an isentropic turbine, equating the entropies at both side of turbine ($s_1=s_2$) gives us the temperature (T_2) and enthalpy (h_2) of $s\text{CO}_2$ at outlet of the turbine. We use equation (8) to find out the mass flow rate of $s\text{CO}_2$
4. We then used an iterative method to find out turbine capacity, initially we assumed our turbine capacity same as our output power we needed i.e., 150 MW
5. We then computed our mass flow rate of $s\text{CO}_2$ needed to produce the desired work
6. Since there is no pressure drop across HTR, pressure at inlet of pre-compressor is same as outlet pressure of turbine ($P_3 = P_2$). Using equation (3) we found out the pressure at outlet of the pre-compressor (P_4)
7. As we have assumed pre-compressor outlet temperature (T_4) to be 167°C and compressor is working isentropically so by equating entropies on both sides ($s_3=s_4$) we get the temperature (T_3) using CoolProp[5]
8. Now, we used literature [1],[3] to find the temperature at which the cooler should cool the $s\text{CO}_2$. It should be more than critical temperature of CO_2 (that is $\sim 31^\circ\text{C}$) so we assumed it to be 32°C . Then we used isentropic condition on main compressor ($s_6=s_7$) and found out the temperature at outlet from the compressor (T_7), using pressure (P_7) that comes from the equation (4) and then using CoolProp[5]
9. Now we used the effectiveness equation (5) on the LTR to find out the enthalpy of $s\text{CO}_2$ (h_5) taking $h(T_7, P_4)$ as a parameter to maximize the denominator as the definition of effectiveness suggests (the fluid achieves maximum transfer rate at those given T and P)
10. Then we found out temperature of $s\text{CO}_2$ at inlet of the cooler (T_5) using CoolProp[4] and found out the enthalpy difference across the cooler assuming no pressure change ($P_5=P_6=P_4$)
11. Using equation (6), we then found out temperature at outlet from the LTR (T_8). Similarly, we used equation (7) to find out the temperature at outlet of HTR (T_9)
12. Then we found out water flow rate to cool down working fluid using equation (9) by assuming the inlet and outlet temperature (T_{10}, T_{11}) and pressure of water (P_{10}, P_{11}).
13. Then we used equation (10) to find out the heat taken out by the cooler. Similarly, using equation (11), we found out heat supplied by the heater to heat the working fluid to the desired value.
14. Now as we know temperature and pressure at all the points, we found out remaining enthalpies using CoolProp[5] and determined the work done by pre-compressor and compressor
15. Then we calculated the net-work output and found that the net-work comes to be very less than we desired. This is because, we considered net-work to be equal to only turbine power output instead of taking net-work to be equal to (turbine power output – pre-compressor power output – compressor power output). Therefore, we increased the value of turbine capacity so that we could achieve the desired output power ($=149.5 \text{ MW}$) and we succeeded at 230 MW turbine.
16. Then we calculated the overall efficiency of the cycle using equation (14).

$$PR_{tur} = PR_{comp} \cdot PR_{precomp} \quad (0)$$

$$P_2 = PR_{tur} \cdot P_1 \quad (1)$$

$$s_2 = s_1 \quad (2)$$

$$P_4 = PR_{precomp} \cdot P_3 \quad (3)$$

$$P_7 = PR_{comp} \cdot P_6 \quad (4)$$

$$\epsilon_{LTR} = \frac{h_4 - h_5}{h_4 - h(T_7, P_4)} \quad (5)$$

$$\dot{m}C_p(T_4 - T_5) = \dot{m}C_p(T_8 - T_7) \quad (6)$$

$$\dot{m}C_p(T_2 - T_3) = \dot{m}C_p(T_9 - T_8) \quad (7)$$

$$\eta_{tur} = \frac{W_{tur}}{\dot{m}_{CO_2}(h_1 - h_2)} \quad (8)$$

$$(\dot{m}_{H_2O})(c_{H_2O})(T_{11} - T_{10}) = \dot{m}_{CO_2}(h_6 - h_5) \quad (9)$$

$$Q_c = \dot{m}_{CO_2}(h_5 - h_6) \quad (10)$$

$$Q_h = \dot{m}_{CO_2}(h_1 - h_9) \quad (11)$$

$$W_{precomp} = \dot{m}_{CO_2}(h_4 - h_3) \quad (12)$$

$$W_{comp} = \dot{m}_{CO_2}(h_7 - h_6) \quad (13)$$

$$\eta_{cycle} = \frac{W_{tur} - (W_{comp} + W_{precomp})}{Q_{heater}} \quad (14)$$

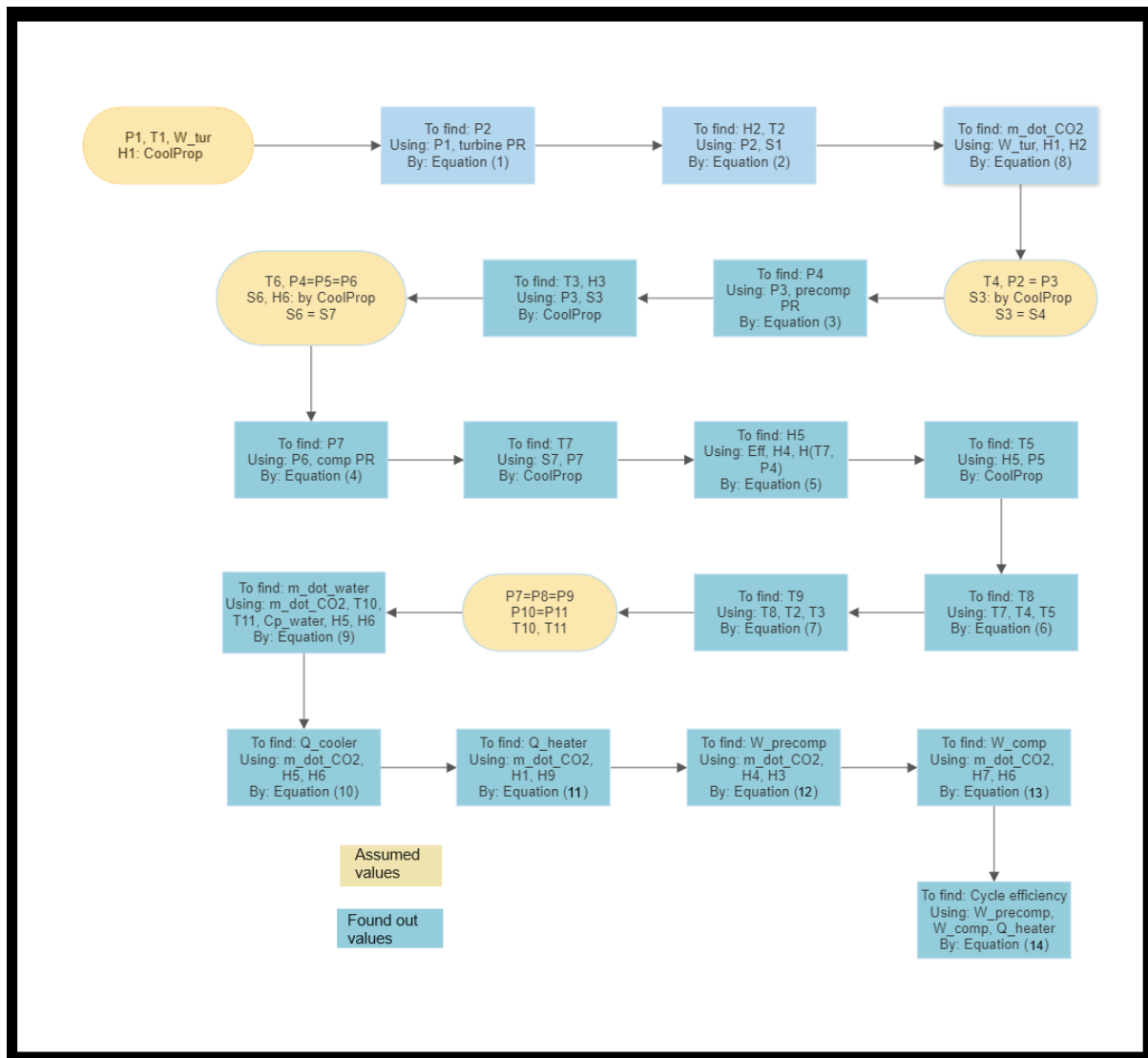


Figure 3: Flow Chart depicting Methodology used

* In the flow chart, H implies specific enthalpy

Component modelling and sizing

Assumptions

The assumptions that we considered to ease our analysis, listed below

- The channels are separated by negligible thickness thus no conductive heat transfer resistance due to that.
- Mean values of temperature are considered to calculate/deduce properties.
- There is negligible pressure drop in channels, no frictional losses, no stress on walls of channel (i.e., forces equate themselves in different directions)

Process of modelling the micro-channel heat exchanger

- By our design specifications we know inlet and outlet temperatures and pressures.

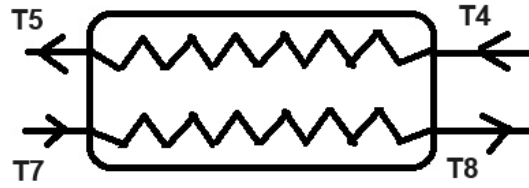


Figure 4. Schematic of LTR

- Started with finding the value of heat load using the enthalpy change in either of the stream and mass flow rate as given in Eq. 15.
- Knowing the inlet and outlet temperatures of the heat exchanger, we found out the Log Mean Temperature Difference (LMTD) Eq. 16.
- Then we got the value of overall heat transfer coefficient(U) times the area needed for the heat exchange(A) Eq. 17

Table 2: List of all design specific parameters.

S. No.	Parameter	Value
1	Temperature inlet hot stream	167 °C
2	Temperature outlet hot stream	66.19 °C
3	Temperature inlet cold stream	51.86 °C
4	Temperature outlet cold stream	159.93 °C
5	Mass flow rate of s-CO ₂	1531.92 kg/sec
6	Channel overall length*	2 m
7	Channel side, s*	1.44 mm

* - Assumed parameters

- Our main aim now was to find out the value of overall heat transfer coefficient, started with finding the value of Raynolds number (Re), using the hydraulic diameter came from Eq. 18
- As we knew the relation Nusselt number (Nu) Eq. 19, Prandtl number being a fluid property we found out using CoolProp. In paper [Ref] author say about the equation[19] is valid for channel size about 1 mm, we used 1.44 mm channel which creates error in our estimation of heat heat transfer coefficient, which comes around 1402 W/m²K, which corresponds to around 1.4x of paer value of heat transfer coefficient used in the paper.
- Another relation for Nusselt number which relates the convective heat transfer coefficient Eq.20, that we needed for knowing overall heat transfer coefficient. Eq. 21
- Using the overall heat transfer coefficient, we found out the area needed for heat exchanger Eq. 17
- Using assumed value of channel side and channel overall length, we found out number of channels per stream (i.e., cold and hot)

Table 3: Derived parameter

S. No.	Parameter	Value
1	Heat load ($\dot{Q}$)	307.625 MW
2	Log mean temperature difference (LMTD)	10.27 °C
3	Average Reynolds number (Re) hot stream	23345.71
4	Average Reynolds number (Re) cold stream	23956.18
5	Average Prandtl number (Pr) hot stream	1.176
6	Average Prandtl number (Pr) cold stream	1.442
7	heat transfer coefficient(h_1) hot stream	2081.59 W/m ² -K
8	heat transfer coefficient(h_2) cold stream	4299.62 W/m ² -K
9	Overall heat transfer coefficient	1402.56W/m ² -K
10	Area of heat exchange	21359.8 m ²

Equation used:

$$\dot{Q} = \dot{m}_{CO_2}(h_4 - h_5) \quad (15)$$

$$LMTD = \frac{(\Delta T_1 - \Delta T_2)}{\log_e \left(\frac{\Delta T_1}{\Delta T_2} \right)} \quad (16)$$

$$\dot{Q} = U \cdot A \cdot LMTD \quad (17)$$

$$Re = \frac{\nu \cdot h_{hyd}}{\mu} \quad (18)$$

$$Nu = 0.023 \cdot (Re)^{0.8} (Pr)^n, n = 0.3 \text{ for hot } 0.4 \text{ for cold} \quad (19)$$

$$Nu = \frac{d_{hyd} \cdot h}{K} \quad (20)$$

$$\frac{1}{U} = \frac{1}{h_{hot}} + \frac{1}{h_{cold}} \quad (21)$$

Cost Functions for all the equipment's

We have used the following cost functions for the turbine, recuperators, cooler, compressors and boilers. We have chosen an axial turbine as our plant is 150 MW and radial turbines are not suited for such high power output plants. We have used the general cost correlation form, given below: $C = a \cdot S^a \cdot P^b \times f_T$, where C is the component cost, a and b are fit coefficients, S is the scaling parameter, and f_T is a temperature correction factor described in the following section. This form has the advantage of adapting to nonlinear and linear cost trends.[10]

Table 4: Cost function for all components.

S.No	Component	Cost Function
1	Turbine	$C_{Tur} = 182600 * \dot{W}_{sh}^{0.5561} * f_{T,Tur}$
2	HTR	$C_{hrecup} = 49.45*(UA)^{0.7544} * f_{T, recup}$
3	Main Compressor	$C_{mcomp} = 6202000 * \dot{V}_{in}^{0.1114}$
4	LTR	$C_{lrecup} = 49.45*(UA)^{0.7544} * f_{T, recup}$
5	Pre-Compressor	$C_{pcomp} = 6202000 * \dot{V}_{in}^{0.1114}$
6	Cooler	$C_{cooler} = 41.17*(UA)^{0.75}$
7	Heater	$C_{heater} = 1248*(\dot{Q})^{6.8071} * f_{T, heater}$

*20% extra cost is accounted for nuts, pipes and labour work along with maintenance cost

Temperature Correction Factor, $f_T = \{ 1 \text{ if } T_{max} < T_{bp} \text{ or } 1 + c(T_{max} + T_{bp}) + d(T_{max} - T_{bp})^2, \text{ if } T_{max} > T_{bp} \}$, where T_{bp} is the temperature breakpoint, typically 550 °C, and T_{max} is the maximum temperature rating of the component, in units of °C.[5] For our system $T_{max} < 550$ °C, always, therefore $f_T = 1$, always.

Results And Discussion

Table 5: Values of power for various components

S.No	Component	Work/Heat Interaction
1	Turbine	230.00 MW
2	Cooler	151.56 MW
3	Heater	311.62 MW
4	Main Compressor	56.44 MW
5	Pre-Compressor	23.22 MW
6	HTR	307.624 MW
7	LTR	289.3866 MW

We obtained the cycle efficiency to be 39.16% which is within the acceptable limit of 59.75%

Parametric Study

The effect of turbine inlet pressure on the cycle efficiency was studied and it was found that as the turbine inlet temperature increases the cycle efficiency also increases. Further, if we increase the turbine efficiency the cycle efficiency increases again. To study the effect of compressor efficiency we fixed the rest parameters and varied the compressor efficiency. It was observed that a higher efficiency naturally contributes higher net work and cycle efficiency at the same HPT inlet temperature. No pressure drops were considered in the Heat exchangers in our study, but in reality, it does not happen, so when we tried to see the effects of pressure drop on the performance of the cycle through our code, it was observed that pressure drop deteriorates the cycle performance. With increase in pressure drop more waste heat cannot be recovered.

Capital Cost Results

Table 6: Cost of various components

S.N.	Component	Calculated Cost (\$)	Uncertainty Range
1	Turbine	3,757,142	25-30%
2	HTR	18,520,225	31-38%
3	Main Compressor	6,722,184	30-50%
4	LTR	21,593,437	31-38%
5	Pre-Compressor	7,649,703	30-50%
6	Cooler	5,180,615	25-28%
7	Heater	633,455	23-26%
Total cost:		76,868,113*	

*20% extra cost is accounted for nuts, pipes and labour work along with maintenance cost

The total cost of our plant came to be 76,868,113 \$ compared with the state of art that we got was around 300 million \$. This huge gap found is due to the heater and cooler. We took a coal powered heater, and the paper used a solar thermal heater.

LTR Modelling

In our power cycle for LTR we choose Micro-channel heat exchanger which consists of multiple very small channels which carry a cold stream in one direction and hot fluid in the other (counter flow). This helps in more heat transfer between hot and cold fluid. We found the overall heat transfer coefficient (U) value from literature [9] which was 1024 W/m²-K. This was for a 100 MW plant. We found out for our plant of 150 MW and U came out to be 1402.56W/m²-K. The channel length and channel side of micro-channel heat exchanger was taken to be 2 m and 1.44 mm respectively [9]. We calculated the heat exchanger area which came out to be 21359.8 m² and the corresponding number of channels came out to be 1,854,153 channels per stream (hot and cold). The heat exchanger dimensions came out to be 2.68m*2.68m*2m. The velocities of the cold and hot streams per came out to be 0.55 m/s and 1.2 m/s respectively.

Conclusion

We analysed a 150 MW sCO₂ power cycle with precompression and recuperation. Using appropriate assumptions and by obtaining values from available literature on sCO₂ power plants, we were able to design a system with some results that matched the typically observed values of efficiency and output work in actual plants. A cost-analysis model was employed to obtain initial capital costs for individual components of the cycle and were further used to draw comparisons with plants having comparable configurations.

References

- [1]. Ing. Martin Kulháněk, Ing. Václav Dostál, SUPERCRITICAL CARBON DIOXIDE CYCLES THERMODYNAMIC ANALYSIS AND COMPARISON
<<https://stc.fs.cvut.cz/history/2009/sbornik/Papers/pdf/KulhanekMartin-319574.pdf>>
- [2]. Anaconda Navigator by Peter Wang and Travis Oliphant, founded on 2012, v2.4.3, last accessed 10-09-23, downloaded from < <https://www.anaconda.com/download> >
- [3]. Colwell, C. P., & Thome, J. R. (2011). Microchannel heat exchangers: A comprehensive review. *Experimental Thermal and Fluid Science*, 36(1-3), 87-122.
- [4]. Ruiyuan Zhang, Wen Su, Xinxing Lin, Naijun Zhou, Li Zhao c Thermodynamic analysis and parametric optimization of a novel S-CO₂ power cycle for the waste heat recovery of internal combustion engines < <https://www.sciencedirect.com/science/article/pii/S0360544220315929> >
- [5]. Ian H. Bell, CoolProp Team 2023, CoolProp, accessed 10 September 2023, <<http://www.coolprop.org/>>
- [6]. Joshua Schmitt, David Amos, Jayanta Kapat *DESIGN AND REAL FLUID MODELLING OF MICRO-CHANNEL RECUPERATORS FOR A NOMINAL 100MW CLASS RECUPERATED RECOMPRESSION BRAYTON CYCLE USING SUPERCRITICAL CARBON DIOXIDE*
- [7]. Smart draw, last accessed 10 September 2023, can be found at < <https://www.smartdraw.com/> >
- [8]. Jiangfeng Guo. *Design analysis of supercritical carbon dioxide recuperator last accessed 11 November 2023*
- [9]. Kehinde Adenuga, Kenneth Means. *Design of a Heat Exchanger for a Supercritical CO₂ Turbine System last accessed 11 November 2023*
- [10]. A study by the National Renewable Energy Laboratory (NREL) titled "*Cost and Performance of Supercritical Carbon Dioxide (SCO₂) Power Plants*" available at <https://www.nrel.gov/docs/fy21osti/77955.pdf>
- [11]. Engineering Toolbox. (n.d.). Carbon Dioxide - Prandtl Number vs. Temperature and Pressure. Retrieved from <https://www.engineeringtoolbox.com/carbon-dioxide-prandtl-number-viscosity-heat-capacity-thermal-conductivity-d_2024.html>